INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
ELECTRICAL ENGINEERING DEPARTMENT

EE-Quiz 2

Friday May 12, 2023	**MS-101 Makerspace**	Marks: 40
Time: 08:15-09:00	Spring Semester 2022-23	(To be re-scaled)

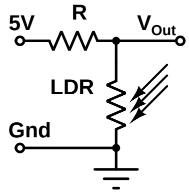
Roll Number: ____________________ **Name:** ____________________________

Division: ______ **Batch:** ________ **Signature:** ____________________________

Q	1	2	3	4	5	6	Total
Marks							

This question paper has 4 pages containing 6 questions. Answer these in the space provided with the questions.
Steps for arriving at numerical answers must be shown.
Numerical answers must be accurate to within 0.1%

Q–1 We use a circuit as shown on the right to detect light using a light dependent resistor (LDR). The resistance of LDR in dark is 100 kΩ. Its resistance when exposed to the brightest light to be measured is 5 kΩ.



a) What should be the value of the series resistor R such that the output voltage V_{Out} is 2.0 V in dark?
Steps:

Answer: __________ – [**2**]

b) If we use a 25 kΩ resistor for R and a rotating wheel with slots to periodically interrupt the brightest light (for which the LDR resistance is 5 kΩ), what will be the peak to peak voltage at the output (V_{Out})?
Steps:

Answer: __________ – [**4**]

c) The rotating wheel used to chop light has 8 slots with equal spacing as shown in part b) above. If the wheel rotates at a constant speed of 150 revolutions per minute, what will be the time period of the periodic signal observed at V_{Out}?
Steps:

Answer: ________ – [3]

– [**Q1: 2 + 4 + 3 = 9 marks**]

Q–2 The voltage output from a sensor ranges between 0 V and 2.0 V. This is to be measured using the in-built 10 bit ADC in Arduino. To maximize resolution of the measurement, we want to use an externally supplied reference voltage of 2.0 V to the ADC.

a) To which pin of Arduino should we connect the reference voltage of 2.0 V?

Answer: ________________ – [2]

b) Which library function should we call and with what argument, so that this reference voltage is used for A to D conversion?

Answer: ________________ – [2]

c) What precaution should we take while writing a sketch using an external reference voltage?

Answer:

– [3]

– [**Q2: 2 + 2 + 3 = 7 marks**]

Q–3 We want to construct a toy car controlled by an Arduino board which can carry out the following four commands:

i) **F** (Move Forward), ii) **B** (Move backwards), iii) **R** (Turn right) and iv) **L** (Turn Left).

Rather than using 4 port pins from Arduino, we want to use just two port pins (D2 and D3) and generate the four commands by decoding the outputs using a digital logic circuit.

The commands are encoded as shown in the table on the right

D2	D3	Command
0	0	**F**: Forward
0	1	**B**: Backwards
1	0	**R**: Turn Right
1	1	**L**: Turn Left

a) Express each of these commands as a logic function of D2, D3 and their complements.

Answer: Fill in the logic function in the table below:

Command		**Logic Function**
F: Forward	=	
B: Backwards	=	
R: Turn Right	=	
L: Turn Left	=	

– [2]

b) We want to generate the decoded commands **F, B, R** and **L** as positive TRUE signals derived from the digital outputs at pins D2 and D3 using **no more than 2 inverters and four 2-input NOR gates** for decoding all commands.

Provide a logic diagram for decoding the four commands from the two digital values (D2 and D3) with this constraint.
Answer:

– [4]

– [**Q3: 2 + 4 = 6 marks**]

Q–4 What will be the value returned by the library function "map" if it is called as: **map(95, 50, 1000, 0, 250)?**

You must provide the steps used for calculating your answer.

Steps:

Answer: __________

– [**Q4: 4 marks**]

Q–5 The library function "attachInterrupt( )" is used for setting up interrupts in an Arduino card. Consider a call to this function as:
attachInterrupt(digitalPinToInterrupt(2), handler, CHANGE);

Subsequent to executing this call,

a) What kind of signal will result in an interrupt?

Answer: ____________________ – [2]

b) On which pin should this signal occur in order to cause an interrupt?

Answer: ____________________ – [2]

c) Which function will be run when an interrupt occurs? Specify its return and argument types.

Answer: ____________________ – [4]

– [**Q5: 2 + 2 + 4 = 8 marks**]

Q–6 Arduino library provides two functions for measuring pulse width: "pulseIn( )" and "pulseInLong( )".

a) Why should interrupts be disabled for an accurate measurement of pulse width if we use the function "pulseIn( )"?

Answer:

– [**3**]

b) Why should interrupts be enabled if we use the function "pulseInLong( )" for pulse width measurement?

Answer:

– [**3**]

– [**Q6: 3 + 3 = 6 marks**]

Paper Ends